

Experiment 21 — FIND-S Algorithm

Aim

To implement the **FIND-S algorithm** to find the most specific hypothesis consistent with a given set of training examples stored in a CSV file.

Algorithm / Procedure

1. Read the training data from the CSV file.
2. Initialize the hypothesis with the most specific values (0).
3. Examine each training example one by one.
4. If the example is **positive (Yes)**, compare it with the current hypothesis.
5. If an attribute differs, generalize that attribute to ?.
6. Ignore negative examples.
7. Display the final most specific hypothesis.

CODE

```
import csv

def find_s(filename):

    with open(filename) as f:

        data = list(csv.reader(f))

        # Initialize hypothesis

        hypothesis = ['0'] * (len(data[0]) - 1)

        # Process training examples
```

```
for row in data:

    if row[-1].strip().lower() == 'yes':

        for i in range(len(hypothesis)):

            if hypothesis[i] == '0':

                hypothesis[i] = row[i]

            elif hypothesis[i] != row[i]:

                hypothesis[i] = '?'

        return hypothesis

print("Most specific hypothesis:", find_s('enjoysport.csv'))
```

OUTPUT

```
... Most specific hypothesis: ['Sunny', 'Warm', '?', 'Strong', '?', '?']
```

Result

The **FIND-S algorithm** was successfully implemented and the most specific hypothesis consistent with the positive training examples was obtained.